

KISANKAVACH

Karnataka

Business Requirements Document

Document control	Value
Document ID	KK-KA-BRD-001
Version	1.0
Status	Product / Delivery Baseline
Pilot geography	Karnataka
Issue date	18 August 2026
Classification	Controlled - programme stakeholders

Farmer Credit. Benefits. Protection.

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Note: Page references are represented by section numbering to preserve stability across approved revisions.

1. Purpose and Scope

Defines the Karnataka pilot baseline for farmer-centred Kisan Credit Card enablement, benefit visibility, consent-driven data exchange, grievance handling, fraud protection and programme oversight.

In scope: farmer mobile experience, assisted service, participating-bank workflow, Government command centre, authorised integrations and operational reporting.

2. Guiding Principles

Kannada-first and low-literacy friendly; reuse verified information; explicit purpose-bound consent; least privilege; interoperability through adapters; configurable workflows and auditability.

The bank remains the final credit decision-maker. Government-originated and bank-originated data retain their source and provenance.

3. Business Context and Outcomes

KCC journeys are fragmented across records, branches and service channels. KisanKavach provides a single guided journey without replacing statutory systems or a bank's credit authority.

Target outcomes: reduced application friction, better completion, transparent status, earlier exception handling, improved SLA visibility and trustworthy benefit/fraud communication.

4. Stakeholders and Ownership

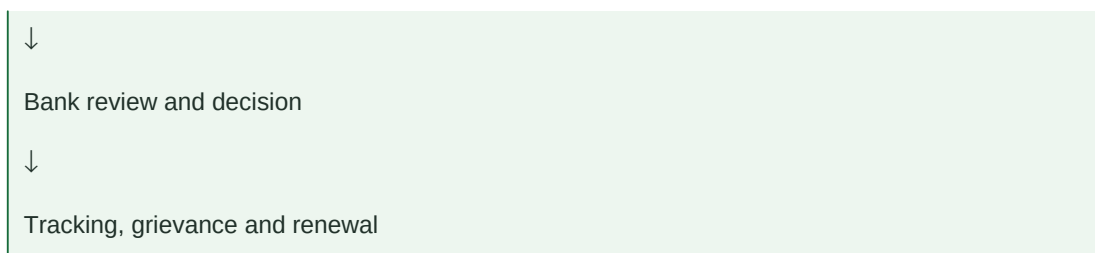
Primary stakeholders are farmers, assisted-service operators, bank officers and managers, Karnataka programme officers, integration owners, security operations and auditors.

Capability	Accountable owner	KisanKavach role
Credit decision	Participating bank	Application orchestration and status
Farmer/land/crop source records	Authorised Karnataka systems	Consent-based retrieval and provenance
Programme oversight	Karnataka Government	Aggregated dashboards and escalation
Platform operations	KisanKavach operator	Availability, security and support

5. Personas and Journeys

Farmer: discover eligibility, prefill verified details, apply, upload, track, respond and complain. Bank officer: receive, verify, request information, assess and decide. Government officer: monitor service performance and systemic issues.





6. Functional Requirements

ID	Requirement	Priority
BR-001	Kannada and English authentication with OTP and secure session controls	Must
BR-002	Consent-based retrieval of farmer, land and crop information	Must
BR-003	Guided KCC eligibility and application wizard	Must
BR-004	Document validation, safe storage and bank requests	Must
BR-005	End-to-end status timeline, alerts and SLA exceptions	Must
BR-006	Bank work queue, appraisal status and decision capture	Must
BR-007	Government dashboards, grievances and programme analytics	Must
BR-008	Benefit and insurance visibility with source labels	Should
BR-009	DigiKavach-aligned fraud warnings and reporting	Should
BR-010	Renewal reminders and repeat application reuse	Should

7. Business Rules

Only verified or farmer-confirmed data may be submitted; every external field records source and retrieval time; one active application per configured product/bank rule; changes after submission are versioned; decisions require structured reason codes; consent denial routes to an allowed manual alternative.

8. Non-Functional and Compliance Needs

Availability target 99.9% for pilot production; responsive farmer interactions under three seconds excluding external latency; encrypted transit and storage; accessible design; complete audit for sensitive actions; retention and deletion aligned to approved policy.

9. KPIs and Acceptance

Measure	Pilot target / interpretation
Application completion	Increase completed submissions against started journeys
Turnaround	Measure median and percentile time by bank/branch/stage
Data reuse	Share of applications using verified prefill
Farmer clarity	Status comprehension and support-contact reduction
Grievance closure	Closure within configured category timelines
Security	No unresolved critical findings at go-live

Go-live requires signed UAT, security approval, operational readiness, partner runbooks, rollback arrangements and accountable owner acceptance.

10. Assumptions, Dependencies and Risks

Dependencies include authorised FRUITS/land/crop access, bank participation, notification providers, policy decisions and support staffing. Key risks are integration delay, data mismatch, digital exclusion, consent confusion, branch adoption and sensitive-data exposure; each requires an owned mitigation and tracked residual risk.

End of controlled baseline

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Software Requirements Specification

Document control	Value
Document ID	KK-KA-SRS-001
Version	1.0
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3. System Context



External authorities remain systems of record. KisanKavach stores references, snapshots and provenance needed for the authorised purpose.

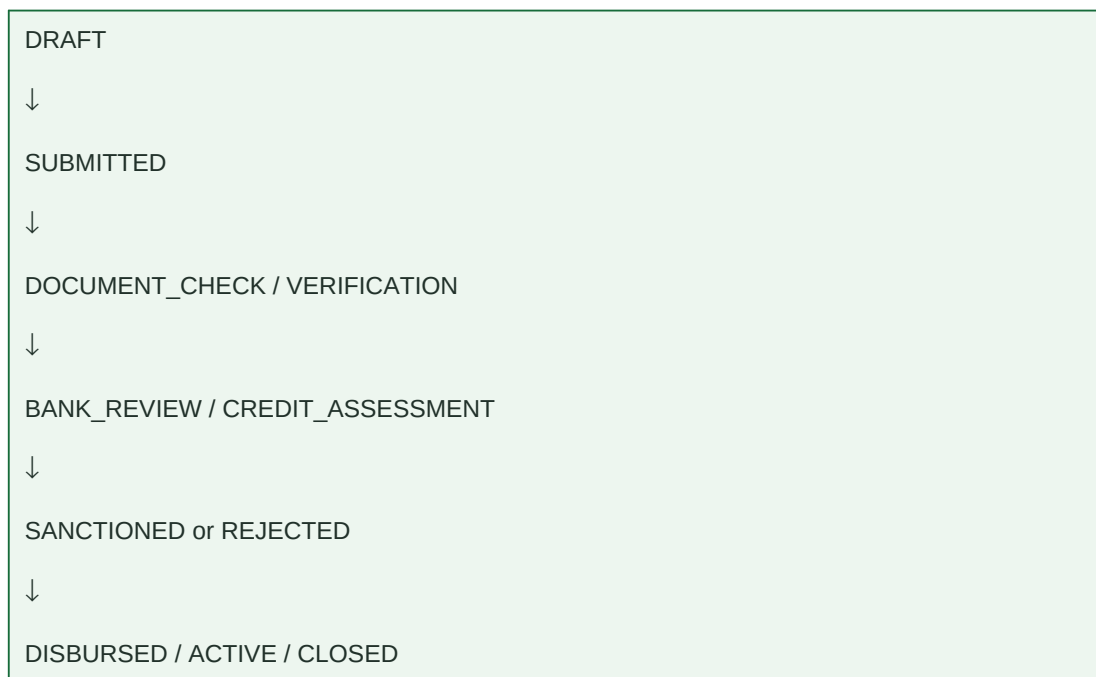
4. Actors and Access

Actor	Permitted scope	Strong controls
Farmer	Own profile, applications, benefits, consents and grievances	OTP, session and device risk
Bank officer	Assigned/branch applications	MFA, role and maker-checker where configured
Government officer	Authorised geography and aggregation level	MFA, scoped RBAC and export control
Administrator	Configuration and operations	Privileged access management and audit
Auditor	Read-only approved evidence	Time-bound access

5. Functional System Requirements

ID	The system shall
FR-001	authenticate supported users and enforce role/scope authorization
FR-002	capture, version, validate and withdraw purpose-specific consent
FR-003	retrieve and label authorised farmer/land/crop data with provenance
FR-004	save application drafts and validate each wizard step
FR-005	submit idempotently and allocate a unique application identifier
FR-006	route applications to the selected bank/branch and start stage timers
FR-007	support document upload, quarantine, malware scan and access control
FR-008	record requests, decisions, reasons, sanction and disbursement events
FR-009	present plain-language timeline and action-required messages
FR-010	create, assign, escalate and close grievances
FR-011	send multilingual, preference-aware notifications
FR-012	produce scoped operational and analytical reports

6. KCC State Model



Every transition requires an allowed source state, authorised actor, timestamp, reason where applicable, correlation ID and immutable audit entry. Repeated inbound events must not create duplicate transitions.

7. Validation and Error Behaviour

Validate at client and server; server rules are authoritative. Errors use stable codes, a farmer-safe message, correlation ID and retry guidance. External timeouts must not lose a draft or create duplicate submissions.

Class	Behaviour
Validation	Highlight field; preserve entries
Authentication	Generic response; throttle and audit
Dependency unavailable	Retry safely; show temporary-unavailability message

Class	Behaviour
Conflict	Return latest state and reconciliation instruction
Security rejection	Block, audit and avoid sensitive detail

8. Non-Functional Requirements

Performance: p95 core reads under 2 seconds and writes under 3 seconds excluding partner delay. Scalability: horizontal stateless services and queued workloads. Resilience: multi-AZ production, backups, tested recovery. Observability: metrics, structured logs, traces and business events without exposed secrets.

9. Data, Privacy and Security

Data minimisation, purpose limitation, encryption, key rotation, secure secrets, masked non-production data, configurable retention, consent evidence, export controls, tamper-resistant audit and incident response are mandatory.

10. Interfaces and Notifications

Support versioned REST/JSON APIs, signed webhooks or controlled batch modes. Kannada and English message templates must identify the application, status and safe next step without exposing sensitive financial or identity data.

11. Acceptance Criteria

Requirements are accepted only when positive, negative, permission, accessibility, resilience and audit tests pass; critical defects are closed; operational dashboards and alerts are demonstrated; and traceability links requirements to test evidence.

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System Architecture

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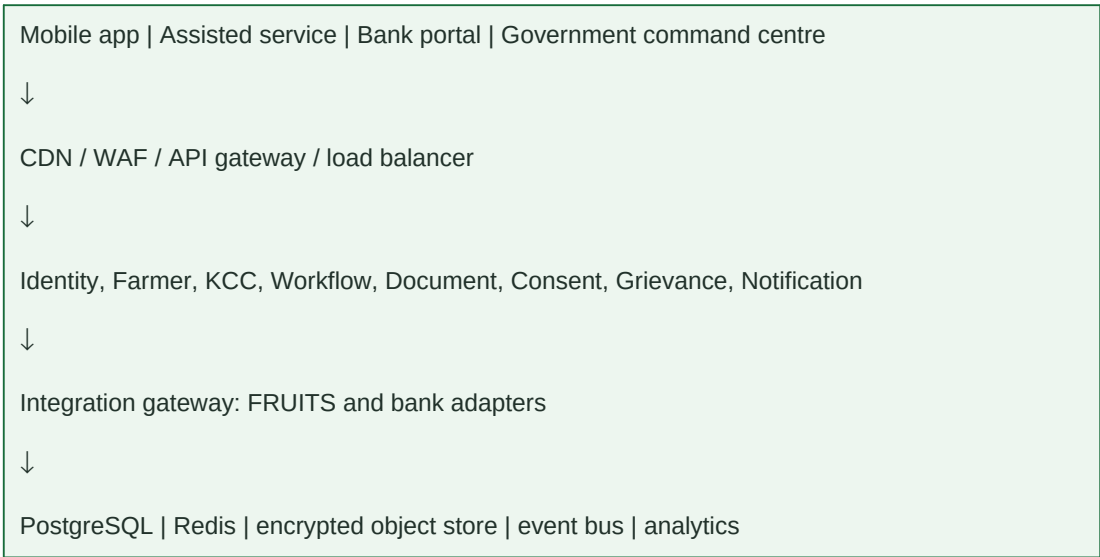
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3. Architecture Overview



4. Deployment and Network

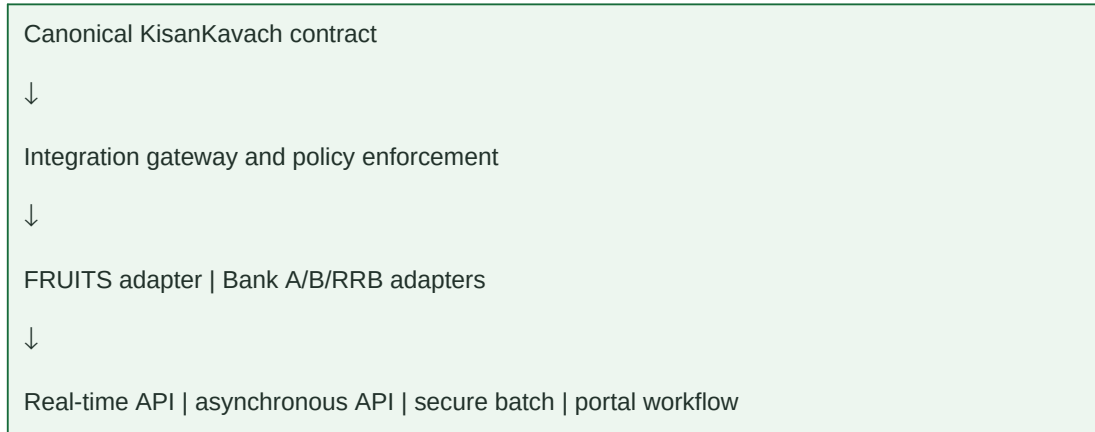
AWS reference deployment uses Route 53, CloudFront, WAF, API Gateway or ALB, private application subnets, isolated data subnets, managed PostgreSQL, Redis, encrypted object storage and protected logs. Development, QA, UAT, pre-production and production are separated.

5. Service Boundaries

Service	Responsibility
Identity	Authentication, tokens, MFA, roles and sessions
Farmer	Profile aggregation, external references and provenance
KCC	Drafts, eligibility guidance, submission, status and renewal
Workflow/SLA	Configurable transitions, timers, alerts and escalation
Document	Metadata, upload, quarantine, scan and retrieval
Consent	Purpose, data categories, recipient, validity and withdrawal

Service	Responsibility
Integration	Canonical contracts, adapters, retry and reconciliation
Audit/Reporting	Evidence, operational metrics and analytical feeds

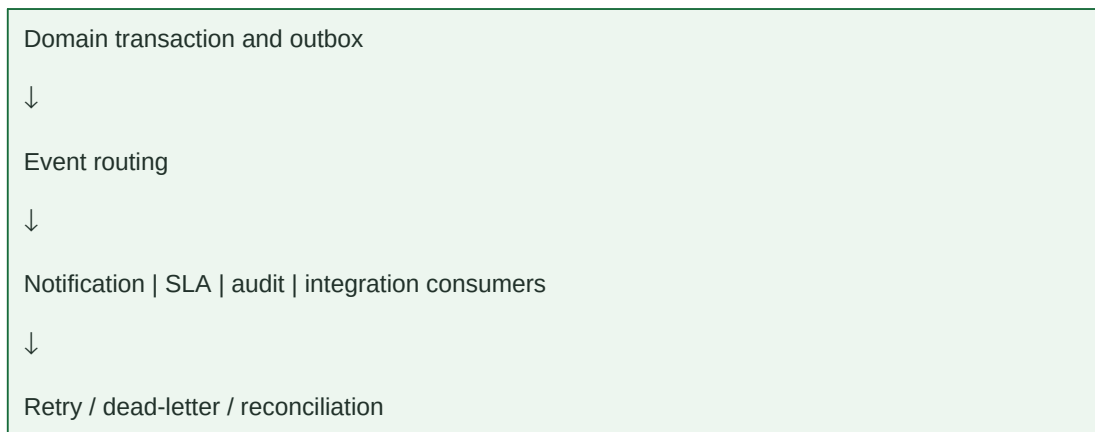
6. Integration Architecture



Adapters isolate core services from partner schema and protocol change. Correlation IDs, idempotency keys, retries with backoff, dead-letter handling and reconciliation are standard.

7. Event and Workflow Architecture

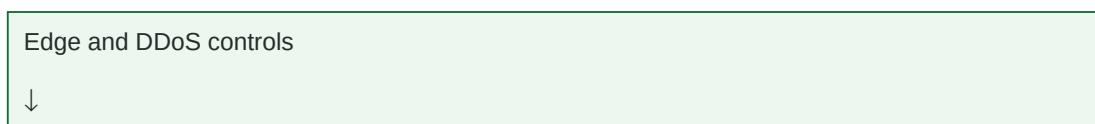
Key events include KCC_APPLICATION_SUBMITTED, DOCUMENT_UPLOADED, BANK_STATUS_CHANGED, KCC_SANCTIONED, KCC_REJECTED, KCC_DISBURSED, SLA_WARNING, SLA_BREACHED, GRIEVANCE_CREATED, FRAUD_DETECTED and CONSENT_WITHDRAWN.

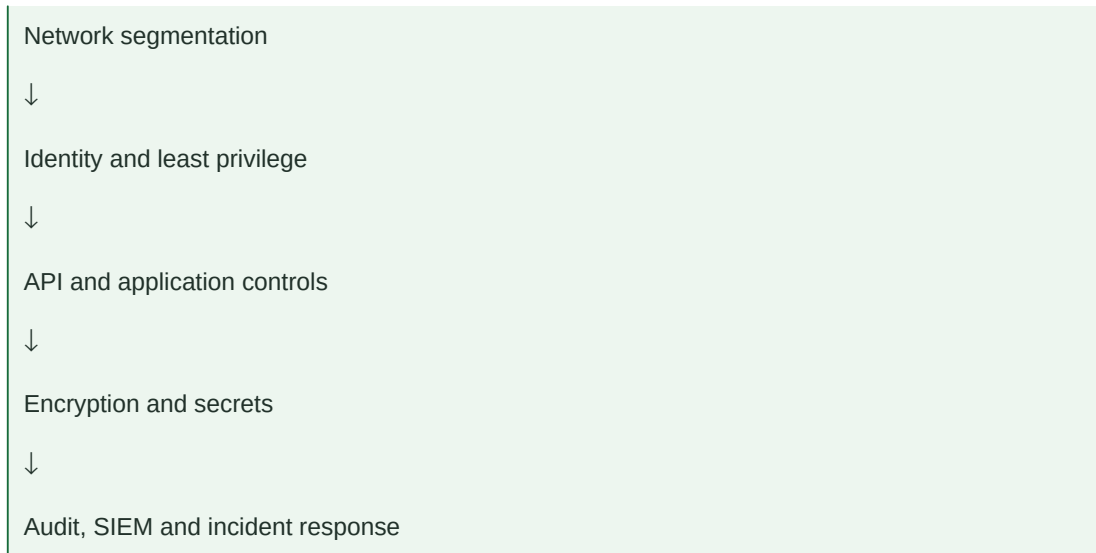


8. Data and Analytics

Operational PostgreSQL is domain-partitioned; Redis is restricted to safe short-lived cache and controls; documents remain in encrypted object storage. An event/ETL pipeline separates Government analytics from transactional workloads.

9. Security Architecture





No public data buckets or Internet-exposed database. Privileged operations require MFA and audit; data exports are scoped, approved and monitored.

10. Availability and Recovery

Concern	Design response
Availability	Multi-AZ managed services and stateless scaling
Failure isolation	Queues, circuit breakers and adapter boundaries
Recovery	Encrypted backups, restore tests, defined RPO/RTO
Deployments	Automated checks, canary/blue-green option and rollback
Observability	SLIs, business metrics, logs, traces and on-call alerts

11. Architecture Decisions

Pilot implementation may use a modular service deployment rather than excessive microservice fragmentation. Canonical APIs and domain boundaries are preserved so high-change or high-scale functions can later be separated.

End of controlled baseline

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Master Database Design

Document control	Value
Document ID	KK-KA-DBD-001
Version	1.0
Status	Product / Delivery Baseline
Pilot geography	Karnataka
Issue date	18 August 2026
Classification	Controlled - programme stakeholders

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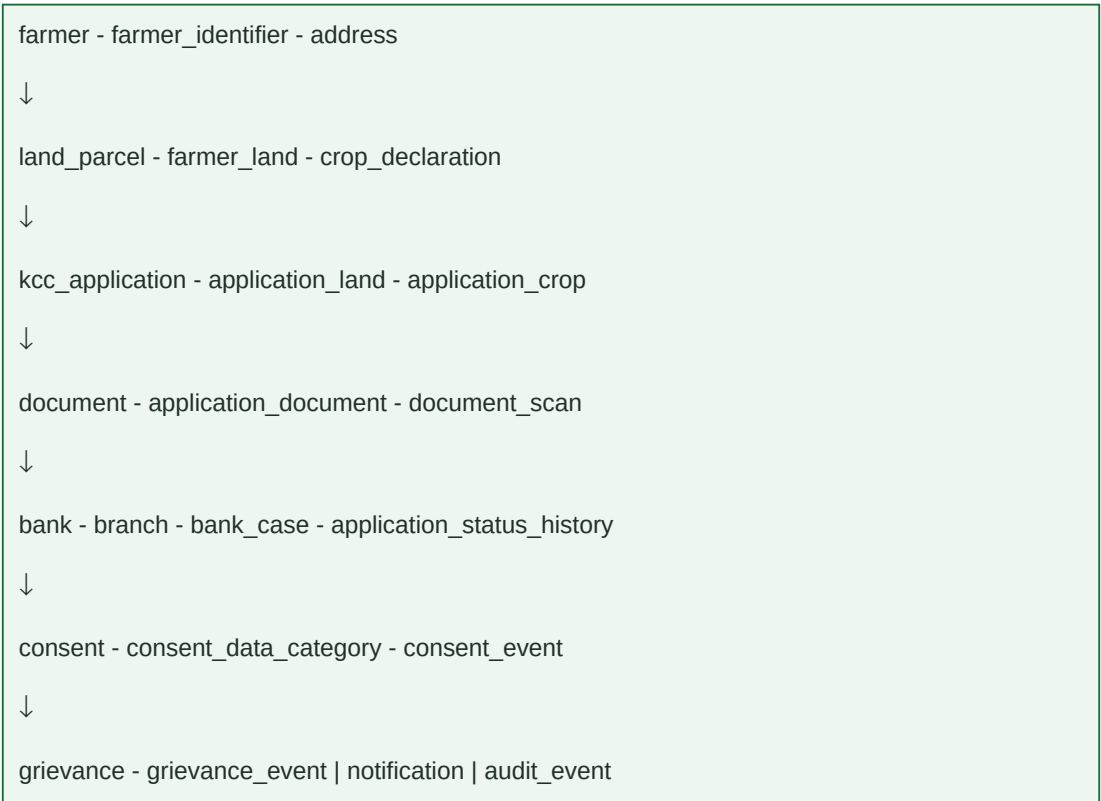
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3. Data Design Principles

PostgreSQL is the system of record for KisanKavach operational state; external Government and bank facts retain provenance. Use UUID/ULID technical keys, separate human references, UTC timestamps, explicit status histories and immutable audit evidence.

4. Logical Domain Model



5. Core Tables

Table	Key fields	Purpose
farmer	farmer_id, display_name, mobile_hash, status	Platform farmer profile

Table	Key fields	Purpose
external_reference	source, reference_value, retrieved_at	Authoritative-system linkage
land_parcel	parcel_id, survey_no, extent, source	Land snapshot and provenance
crop_declaration	crop_id, season, area, source, status	Crop evidence
kcc_application	application_id, public_ref, bank_id, amount, status	KCC aggregate root
application_status_history	from_status, to_status, actor, occurred_at	Immutable lifecycle
document	document_id, object_key, checksum, scan_status	Secure file metadata
consent	purpose, recipient, granted_at, expires_at, status	Consent evidence
grievance	category, priority, owner, due_at, status	Case management
audit_event	actor, action, resource, correlation_id, event_at	Protected evidence

6. Application Relationships

A farmer may have multiple applications over time. Each application snapshots selected land/crop and verified values to preserve the submitted record. Status is materialised on the application for fast reads and fully represented in append-only history.

7. Status and Reference Data

Entity	Representative controlled values
Application	DRAFT, SUBMITTED, DOCUMENT_CHECK, BANK_REVIEW, SANCTIONED, REJECTED, DISBURSED, ACTIVE, CLOSED
Consent	REQUESTED, GRANTED, DENIED, EXPIRED, WITHDRAWN
Document scan	QUARANTINED, SCANNING, CLEAN, REJECTED
Grievance	OPEN, ASSIGNED, IN_PROGRESS, RESOLVED, CLOSED, REOPENED

8. Indexes, Constraints and Concurrency

Unique public application reference; composite indexes on bank/branch/status/updated_at and farmer/status; foreign keys for ownership; check constraints for positive amount/extent; optimistic version column on mutable aggregates; partial unique rules for active applications where policy requires.

9. Security and Privacy

Separate sensitive identifiers from display data; encrypt high-risk columns where needed; use row/scope enforcement in services; restrict raw production access; log reads and exports; tokenize or hash search identifiers; never store secrets in tables.

10. Retention, Partitioning and Archival

Partition high-volume audit, notification and status-event tables by time. Retention is category- and purpose-specific. Expiry jobs create auditable disposition records; legal or regulatory holds override deletion.

11. Migration and Quality

Schema changes are versioned, backward compatible during rolling deployment, reviewed and tested on production-like volumes. Data-quality checks cover duplicates, orphan references, invalid transitions, missing provenance and consent/application inconsistencies.

12. Backup and Recovery

Encrypted automated backups and point-in-time recovery are required. Restore drills validate integrity, application compatibility and documented RPO/RTO. Backup access and deletion are tightly restricted and audited.

End of controlled baseline

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API & Integration Specification

Document control	Value
Document ID	KK-KA-API-001
Version	1.0
Status	Product / Delivery Baseline
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3. API Standards

HTTPS REST/JSON under /api/v1; UTF-8; ISO-8601 UTC timestamps; Indian currency in integer paise or documented decimal; stable enums; OpenAPI contract; correlation ID on every request; no sensitive data in URLs.

4. Authentication and Headers

Header	Requirement
Authorization	Bearer token for protected calls
X-Correlation-ID	Client supplied or generated and returned
Idempotency-Key	Required for submission and payment-like mutations
Content-Type	application/json or approved multipart upload
X-Consent-ID	Required when the operation relies on farmer consent

5. Core Endpoints

Method / path	Purpose	Key response
POST /auth/otp/request	Request OTP	challenge_id, expiry
POST /auth/otp/verify	Verify and create session	access/refresh token metadata
GET /farmers/me	Get aggregated profile	profile plus provenance
POST /consents	Record decision	consent_id and status
POST /kcc/applications	Create draft	application_id/public_ref
PATCH /kcc/applications/{id}	Save validated draft step	version and validation result
POST /kcc/applications/{id}/submit	Idempotent submission	status and submitted_at
GET /kcc/applications/{id}/timeline	Farmer-safe status history	current state and actions
POST /documents/uploads	Initiate secure upload	upload token/object reference
POST /grievances	Create grievance	case reference and due date

6. Canonical Payload

```
{
  "applicationId": "...",
  "farmerId": "...",
  "bank": {"bankId": "...", "branchId": "..."},
  "requestedAmount": {"currency": "INR", "value": "200000.00"},
  "consentId": "...",
  "version": 4
}
```

7. Errors and Idempotency

HTTP	Code	Behaviour
400	VALIDATION_FAILED	Field errors; do not retry unchanged
401/403	AUTH_REQUIRED / FORBIDDEN	Reauthenticate or deny without leakage
409	STATE_CONFLICT	Refresh current resource/version
422	BUSINESS_RULE_FAILED	Explain allowed corrective action
429	RATE_LIMITED	Retry after advised interval
503	DEPENDENCY_UNAVAILABLE	Safe retry with backoff

An idempotency record binds key, caller, operation and request hash; repeats return the original outcome, while a different payload using the same key is rejected.

8. FRUITS and Government Integration

Adapter maps authorised farmer, land and crop responses into canonical objects. Every response records source system, source reference, retrieval timestamp, verification status and consent ID. Cache only within approved purpose and freshness rules.

9. Bank Integration Contract

Common operations: submitApplication, getApplicationStatus, requestAdditionalDocument, updateSanctionStatus, updateRejectionStatus, updateDisbursementStatus and getRenewalStatus. Supported modes are real-time REST, secure asynchronous API, secure batch/file exchange and controlled portal workflow.

10. Webhooks and Events

Signed callbacks include event_id, event_type, occurred_at, resource reference, sequence/version and correlation ID. Consumers verify signature, timestamp and replay window, persist receipt, process idempotently and return promptly.

11. Operational Controls

Per-partner credentials and limits; timeouts; exponential backoff with jitter; circuit breakers; dead-letter queues; dashboards; reconciliation reports; certificate/key rotation; sandbox certification and production cutover checklist.

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QA, Testing, UAT & Requirements Traceability Matrix

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3. Quality Strategy

Risk-based testing covers functionality, integration, permissions, security, privacy, accessibility, performance, resilience, data migration and operations. Test evidence must be reproducible, environment-labelled and linked to build and requirement.

4. Test Levels

Level	Focus	Exit evidence
Unit/component	Rules, validators, adapters	Automated coverage and clean build
API/service	Contracts, state, idempotency, authorization	Contract suite and negative cases
System/E2E	Farmer, bank and Government journeys	Scenario evidence
Integration	FRUITS, banks, SMS/push, documents	Partner/simulator certification
Security/privacy	OWASP, access, consent, audit, leakage	No open critical/high risk
Performance/resilience	Load, soak, failover, recovery	SLO and recovery results
UAT	Business fitness and usability	Signed acceptance

5. Environment and Data

QA and UAT use synthetic or approved masked data. Test identities represent every role and geographic scope. External simulators support success, latency, timeout, malformed response, duplicate callback and partial failure.

6. Priority Test Scenarios

ID	Scenario	Expected result
TC-001	Kannada OTP login and masked mobile	Session created; safe display and audit
TC-002	Consent grant, retrieval and withdrawal	Access follows consent state and purpose
TC-003	Prefilled land/crop with provenance	Source and timestamp retained
TC-004	Draft save and resume on poor network	No loss or duplicate submission

ID	Scenario	Expected result
TC-005	Malicious document upload	Quarantined, blocked and security event
TC-006	Bank requests additional document	Farmer action and timeline update
TC-007	Sanction/rejection callback replay	Single valid transition
TC-008	District user requests other district	Denied and audited
TC-009	SLA breach	Escalation and dashboard reflected
TC-010	Dependency outage during submit	Safe retry and reconciled outcome

7. UAT Plan

UAT participants include farmer representatives, assisted operators, branch and regional bank users, programme officers and support teams. Sessions cover happy paths, exceptions, Kannada comprehension, accessibility, operational reports and incident runbooks.

8. Defect Management

Severity	Definition	Release rule
S1 Critical	Security exposure, data loss, system unavailable	Zero open
S2 High	Major journey blocked; no acceptable workaround	Zero open unless formal risk acceptance
S3 Medium	Impaired function with workaround	Owned plan and acceptance
S4 Low	Minor cosmetic/documentation issue	May defer by agreement

9. Requirements Traceability Matrix

Requirement	Design/API	Tests	UAT
BR-001 / FR-001	Identity, OTP endpoints	TC-001 + security	Farmer login
BR-002 / FR-002-003	Consent/Farmer services	TC-002-003	Consent clarity
BR-003 / FR-004-006	KCC/workflow	TC-004, TC-010	Application journey
BR-004 / FR-007	Document service	TC-005	Upload and bank request
BR-005 / FR-008-009	Timeline/SLA/events	TC-006-009	Status comprehension
BR-006 / FR-008	Bank portal/adaptor	TC-006-007	Officer workflow
BR-007 / FR-010-012	Grievance/reporting	Scope and SLA suite	Programme oversight

10. Entry, Exit and Sign-off

Entry: stable build, approved scope, test data, available environments and interfaces. Exit: planned execution complete, regression pass, performance and security approval, reconciled evidence, no prohibited open defects, operational readiness and signed UAT.

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UI/UX & Screen Specification

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3. Experience Principles

Farmer first; one primary action per screen; plain language; status before detail; progressive disclosure; source visibility; accessibility; Kannada and English with language change at any time.

A farmer should always know what they have, what they need to do, where their KCC application stands, and where to get help.

4. Navigation and Design System

Farmer bottom navigation: HOME | KCC | BENEFITS | HELP | MORE. Use 44px minimum targets, strong contrast, visible focus, meaningful icons with labels, Indian number formatting, low-bandwidth states and no colour-only status meaning.

Token	Use
Kavach green	Primary actions, verified state, headers
Amber	Action required or delay
Red	Critical fraud/security warning only
Neutral grey	Unavailable/inactive information
Typography	Readable sans serif; Kannada-capable production font

5. Priority Screen Catalogue

ID	Screen	Primary outcome
KK-AUTH-002	Language selection	Persist Kannada/English
KK-AUTH-004/005	Mobile and OTP	Secure access
KK-FRM-001	My Kisan Home	Status, action and alerts
KK-CNS-001	Consent	Purpose and data choice
KK-KCC-002/003	Eligibility	Guidance, not sanction promise
KK-KCC-004-011	Application wizard	Farmer to review in nine steps
KK-KCC-009	Documents	Verified vs required
KK-KCC-012	Submission	Reference and next step

ID	Screen	Primary outcome
KK-KCC-013	Timeline	Current state and action
KK-KCC-014/015	Decision	Sanction or respectful rejection
KK-GRV-001	Grievance	Category, evidence and reference
KK-BNK-001	Bank dashboard	Queue and SLA
KK-GOV-001	Command centre	Programme overview

6. Farmer Home Wireframe

KisanKavach | Namaskara, Farmer

ACTION REQUIRED - 1 document pending [Upload Now]

MY KCC - In Bank Review [View Status]

MY BENEFITS - PM-KISAN Active | Crop Insurance Active

KAVACH ALERTS - No high-risk alerts

HOME | KCC | BENEFITS | HELP | MORE

Critical fraud warnings appear first; then required actions; then KCC and benefits. Empty states explain what is absent and offer a single useful next action.

7. Application Wizard

1 Farmer



2 Land



3 Crop



4 Bank



5 Loan requirement



6 Documents



7 Consent



8 Review

↓

9 Submit

Verified data is read-only or clearly editable through an issue path. Save progress automatically. Each step states why information is needed, validates inline and preserves previous entries.

8. Application Status and Decision

Timeline: Submitted - Documents verified - Bank review - Credit assessment - Decision - Disbursement. Current state is highlighted and accompanied by 'Do you need to do anything?'. Avoid the term SLA for farmers; say 'taking longer than expected'.

9. Bank Portal

Area	Contents
Dashboard	Assigned queue, action required, ageing and SLA
Application	Farmer/land/crop with provenance and consent
Documents	Safe preview, verification and request action
Decision	Structured sanction/rejection, reason and maker-checker
Reports	Branch workload, turnaround and exceptions

10. Government Command Centre

State, district, taluk and bank views show volumes, conversion, ageing, SLA breaches, grievances and fraud signals. Default to aggregate data; record-level access requires explicit scope and business need. Filters and exports preserve applied scope.

11. States, Content and Accessibility

State	Required treatment
Loading	Progress without blocking unrelated content
Empty	Explain and offer next step
Offline	Preserve draft and show sync status
Error	Plain message, retry/help, correlation reference
Success	Confirm result, reference and next action
Permission denied	No data leakage; route to support if appropriate

All screens support screen readers, logical focus, scalable text, keyboard portal access, clear validation, alt text and bilingual content review.

12. Screen Acceptance Checklist

Correct role and scope; primary action obvious; Kannada and English reviewed; states and validation specified; API and analytics events mapped; privacy notice and consent correct; accessibility passed; audit and error behaviour confirmed.

End of controlled baseline